

universität freiburg

**DESIGN COURSE: RF- AND MICROWAVE
CIRCUITS AND SYSTEMS**

BLOCK 1

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PE1

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Task2

1. Number 2.) In theory: Independent of its length. It is only determined by the geometry(a, b) and material parameters (L, C)

$$Z_0 = \sqrt{\frac{L}{C}}$$

In practice the Z_0 is still independent of the length however it is depended on the frequency of the signal (ω)

$$Z_0 = \sqrt{(R + j\omega L)/(G + j\omega C)}$$

2. Number 3.)Yes. Electrical length is frequency dependent.
3. Number 5:)

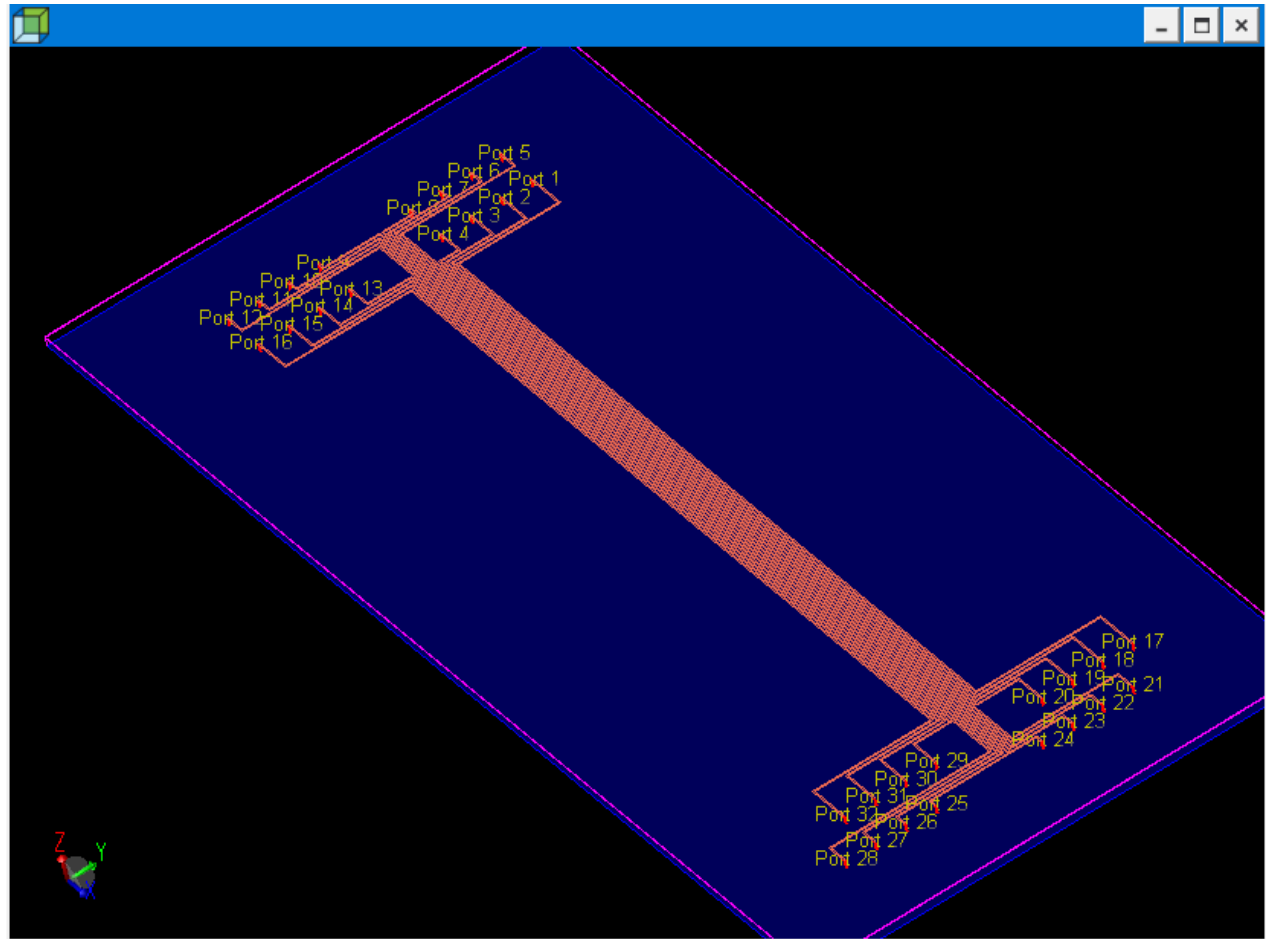
$$Z_0 = \frac{60}{\sqrt{\epsilon_r}} \ln \left(\frac{b}{a} \right)$$

- $D_o = 6.66 \text{ mm}$

4. Number 6 :)Ground to Ground Spacing is $2W + G = 16.08 \mu m$
5. Number 8:) $70 \Omega: W = 30 \mu m$
 $30 \Omega: W = 178 \mu m$
-

Task3

1. Number2:)



S-parameter Ports

Number	Gnd Layer	Name	Feed Type	Ref Impedance [Ohm]	Ref Offset [um]	Term Type
1	<Implicit>	P1	Direct	50 + 0i	N/A	inputOutput
+ P1 ● Gnd						
2	<Implicit>	P2	Direct	50 + 0i	N/A	inputOutput
+ P2 ● Gnd						
3	<Implicit>	P3	Direct	50 + 0i	N/A	inputOutput
+ P3 ● Gnd						
4	<Implicit>	P4	Direct	50 + 0i	N/A	inputOutput
+ P4						

Layout Pins

Name	Layer	Net	Connected to	Purpose	X [um]	Y [um]	Number	L
P1	cond	N_15	1(+)	drawing	-65989.2	-8255	1	1
P2	cond	N_14	2(+)	drawing	-65989.2	-13335	2	1
P3	cond	N_13	3(+)	drawing	-65989.2	-18415	3	1
P4	cond	N_12	4(+)	drawing	-65989.2	-23495	4	1
P5	cond	N_11	5(+)	drawing	-73609.2	-8255	5	1
P6	cond	N_10	6(+)	drawing	-73609.2	-13335	6	1
P7	cond	N_9	7(+)	drawing	-73609.2	-18415	7	1
P8	cond	N_8	8(+)	drawing	-73609.2	-23495	8	1
P9	cond	N_7	9(+)	drawing	-73660	-38735	9	1
P10	cond	N_6	10(+)	drawing	-73660	-43815	10	1
P11	cond	N_5	11(+)	drawing	-73660	-48895	11	1
P12	cond	N_4	12(+)	drawing	-73660	-53975	12	1
P13	cond	N_3	13(+)	drawing	-66040	-38735	13	1
P14	cond	N_2	14(+)	drawing	-66040	-43815	14	1
P15	cond	N_1	15(+)	drawing	-66040	-48895	15	1
P16	cond	N_0	16(+)	drawing	-66040	-53975	16	1

2. Number3:)

- Linecalc only calculates transmission line parameters
- Momentum performs full EM field simulation
- Structures: bends, junctions or coupled lines

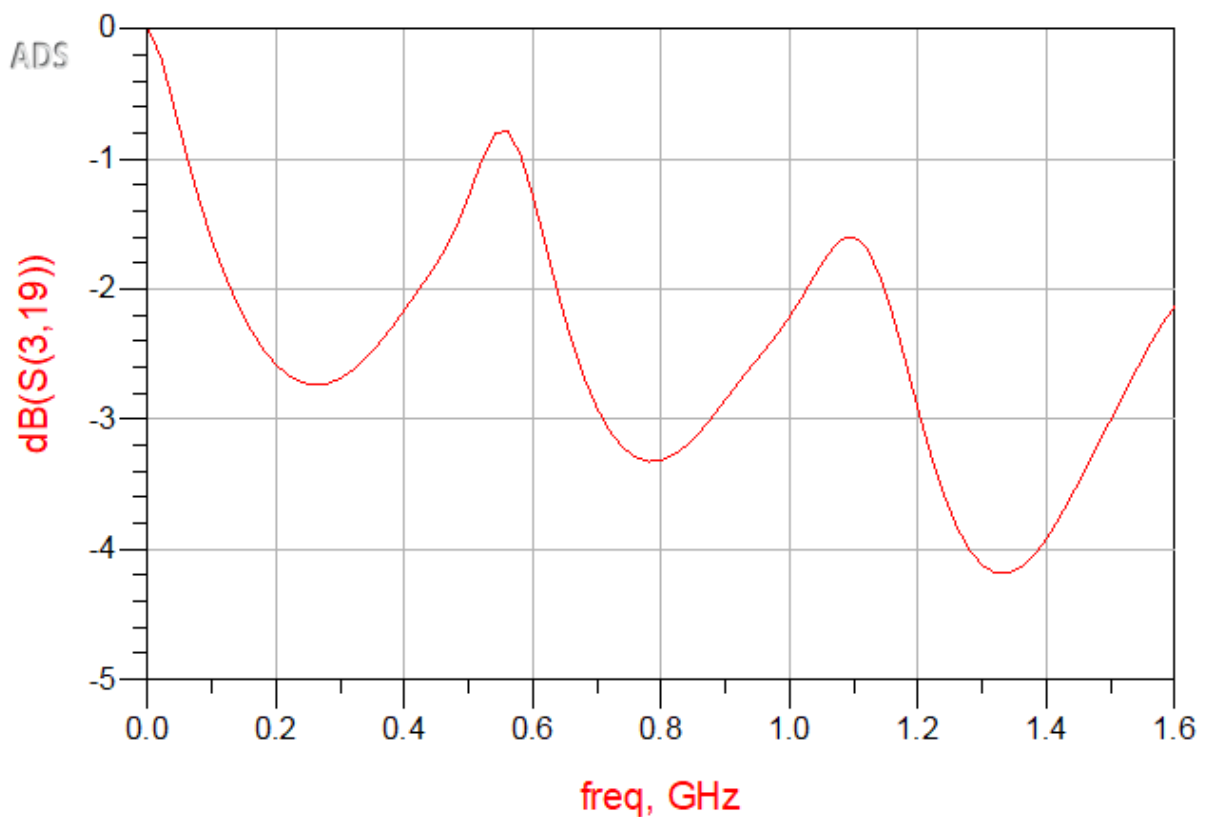
3. Number4:)

A transmission line transforms the load impedance depending on its characteristic impedance and electrical length.

A transmission line introduces a phase shift to the traveling wave proportional to the line length and wavelength.

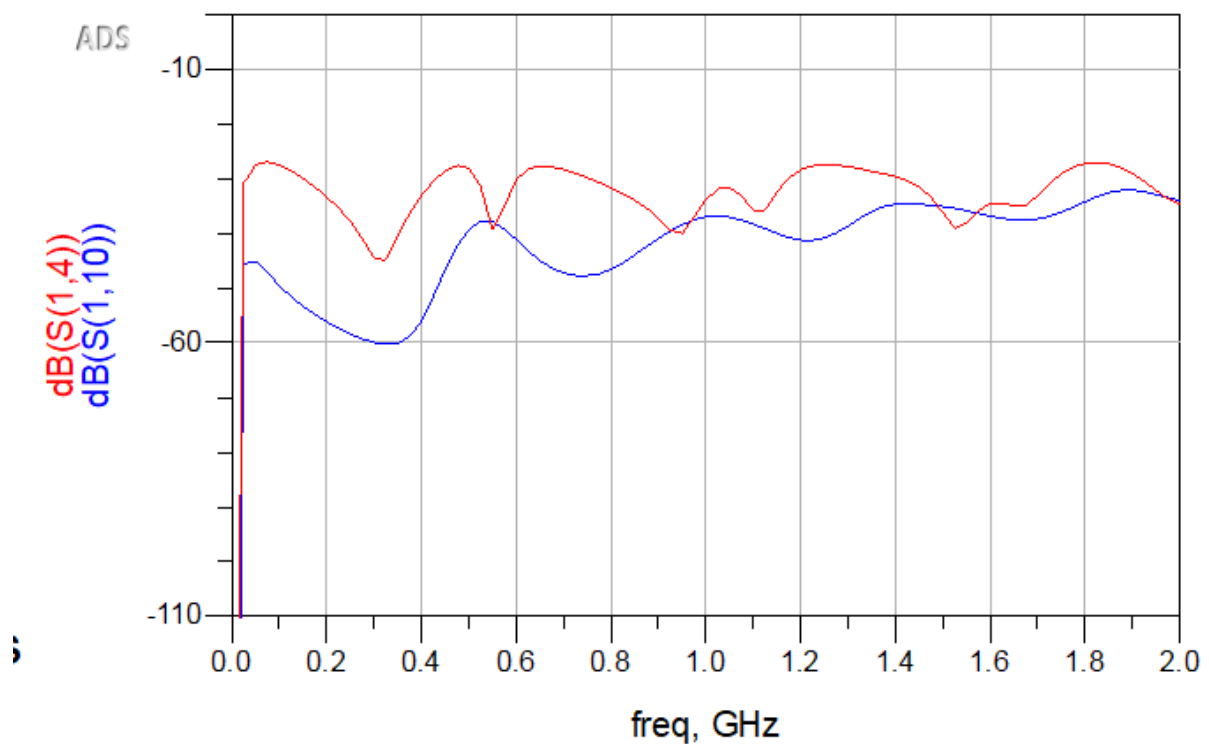
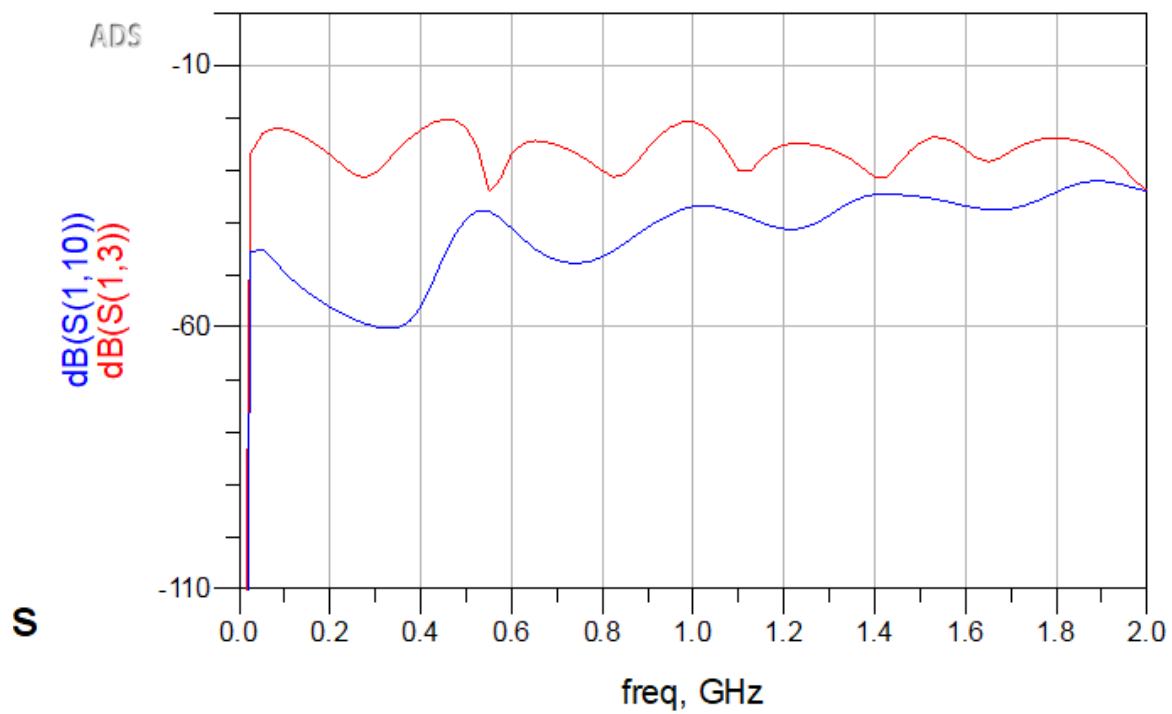
A transmission line can cause reflections if the load impedance does not match the characteristic impedance of the line.

4. Number5:)



- The resonating behavior is caused by standing waves due to reflections along the transmission lines
- Line length is a multiple of $\lambda/2$ (maxima)
- $\lambda/4$ (minima)

5. Number7:)



- $S(1,3)$ shows stronger crosstalk because the lines are physically closer and have stronger electromagnetic coupling
- $S(1,10)$ is weaker due to larger distance

6. Number8:)

$$P_{out} = P_{in} \cdot |S(10,1)|^2$$

$$S(10,1)_{max} = -32.15 \text{ dB}$$

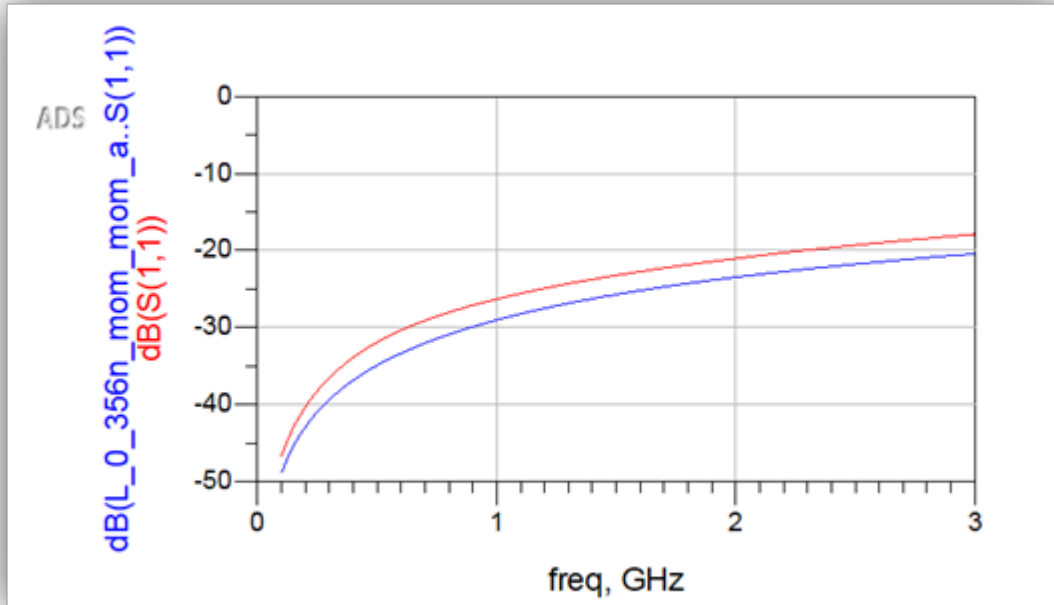
$$P_{out} = 6.1 \text{ mW}$$

7. Number9:)

- directional coupler
- high-speed circuit

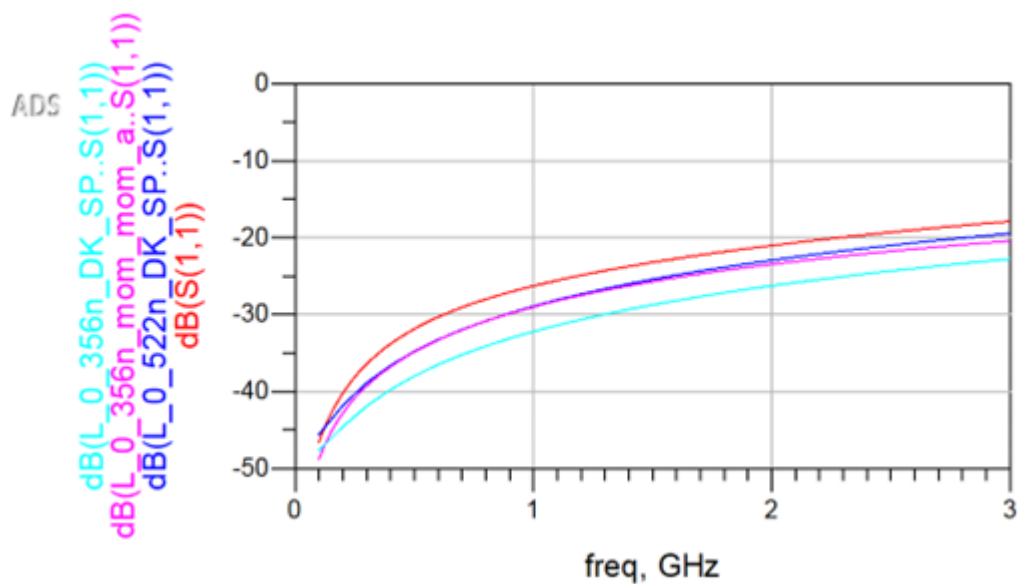
Task4

1. Number 3:)



- A pure inductor shows an impedance increasing with frequency.
- Ideally, the inductor has no resistive loss or parasitic capacitance.
- The phase response should remain close to +90 degrees.

2. Number 4:)

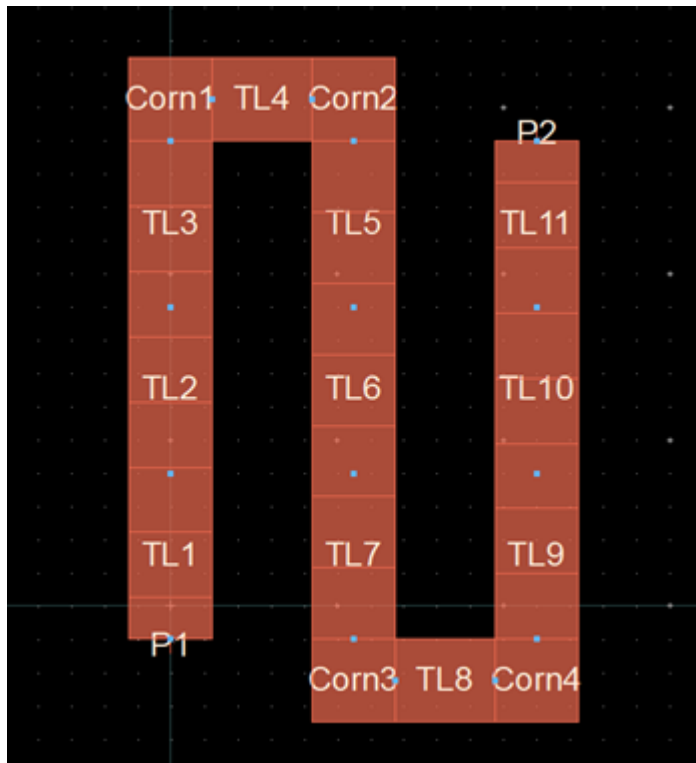


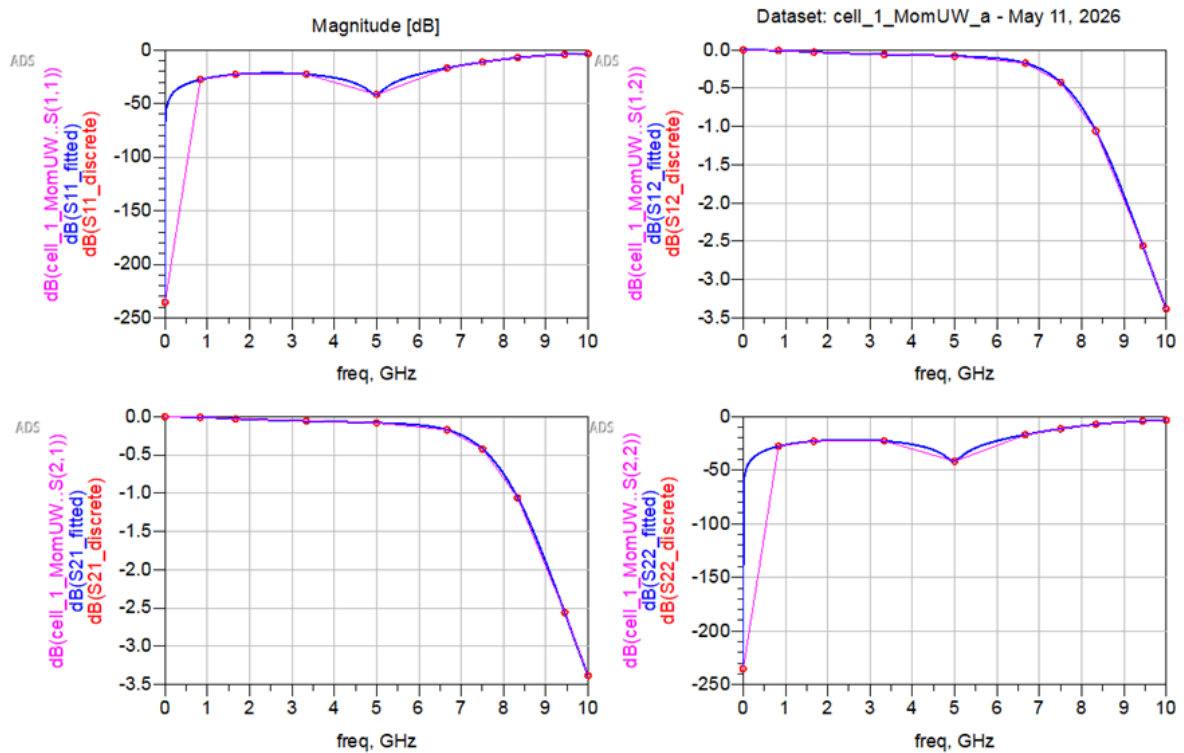
- The Momentum simulation deviates from the compact model at higher frequencies.
- The EM simulation shows additional loss and parasitic effects.
- The compact model behaves more ideally than the full electromagnetic simulation.

3. Number5:)

- The layout introduces parasitic capacitances and coupling effects.
- Metal losses and substrate losses are included in the Momentum simulation.
- The compact model is simplified and cannot fully represent distributed electromagnetic behavior.

4. Number6:)





- The Momentum simulation shows higher losses than the schematic simulation.
- The bends and discontinuities introduce additional parasitic effects.
- Electromagnetic coupling and radiation are only captured in the EM simulation.

Task5

1. Number 2:)

4 ports

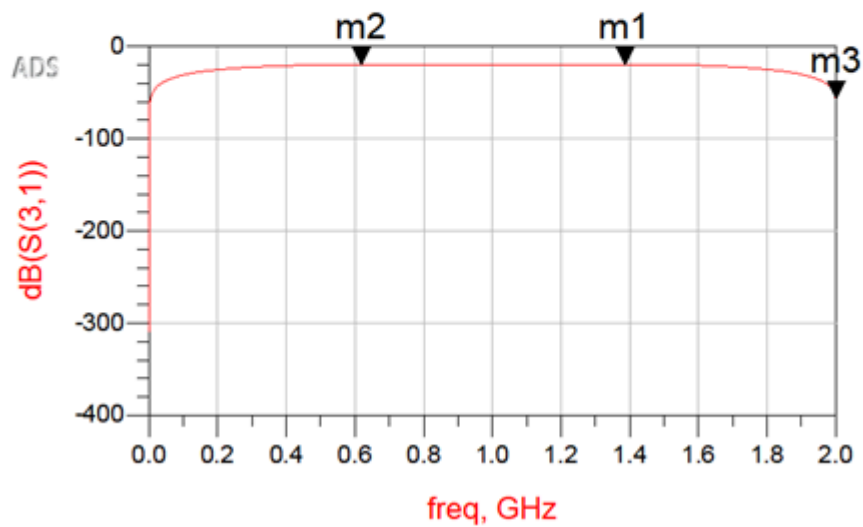
- Port1: Input
 - Port2: Through
 - Port3: Coupled
 - Port4: Isolated
- Quantities
- Coupling
 - Insertion loss
 - Directivity

2. Number3:)

m1
freq=1.385 GHz
dB(S(3,1))=-19.817
Peak

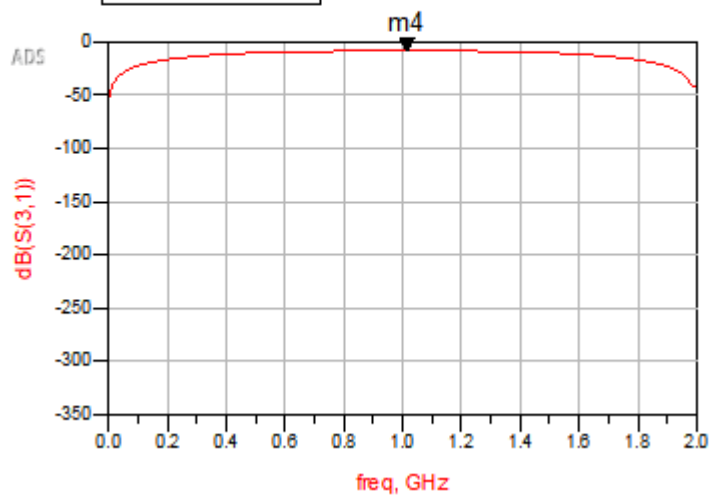
m2
freq=618.1 MHz
dB(S(3,1))=-20.121
Peak

m3
freq=2.000 GHz
dB(S(3,1))=-56.669
Valley

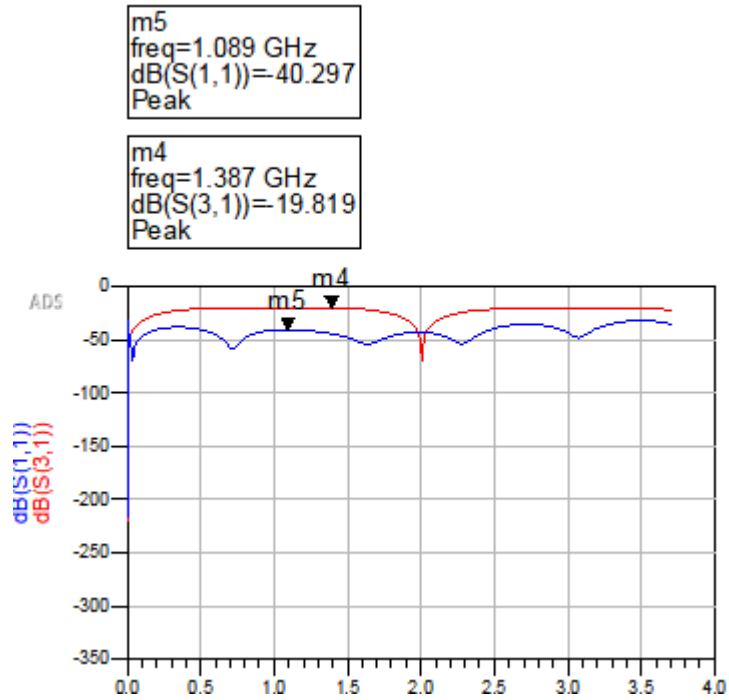


Maximal we can get -8db

m4
freq=1.014 GHz
dB(S(3,1))=-8.301
Peak



3. Number4:)



4. Number5:)

